



MIE1624

Group Project Report

Course Instructor:
Prof. Oleksandr Romanko

A Vision for Canada's Future Innovation

Canada's Innovation Ecosystem Development Analysis & Strategy

”

Engaged pro bono by the University of Toronto, our team evaluates Canada's innovation global landscape and proposes practical steps for strategy implementation.

Group 8

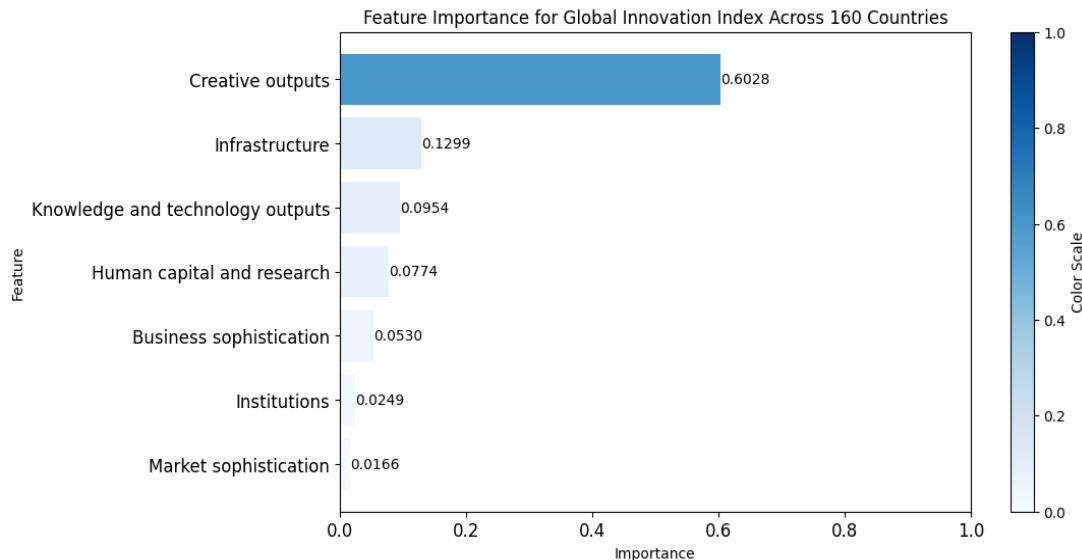
Danni Yang
Jie Zhao
Ruojin Lin
Sarah Khairallah
Si Cheng
Yuchen Zhang
Yueran Yang

Global Innovation Index (GII): Unveiling Key Features

Essential for understanding the innovation landscape globally. Utilizing Random Forest Regression, revealed Creative Outputs significantly impacts the GII, followed by Infrastructure and Knowledge and Technology Outputs.

Framework for GII:

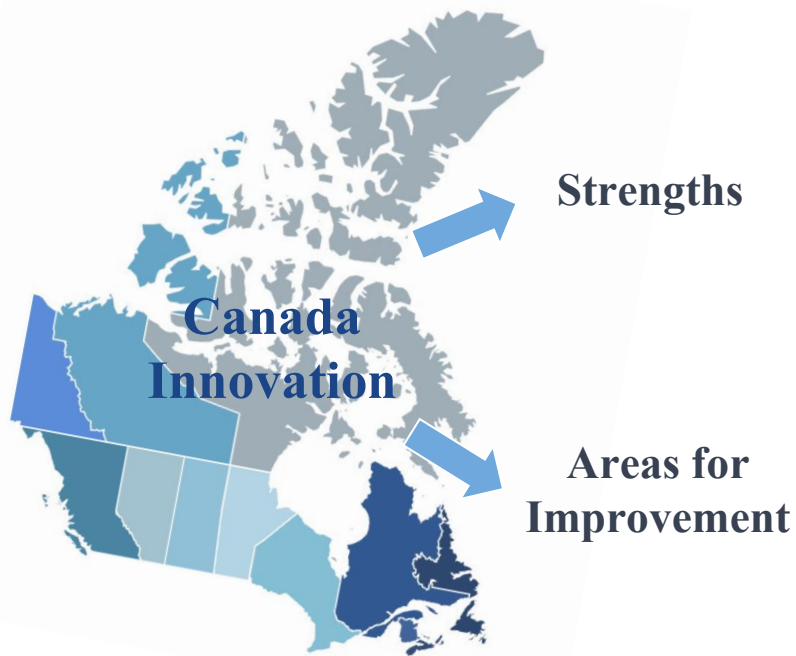
- Institutions
- Human Capital & Research
- Infrastructure
- Market Sophistication
- Business Sophistication
- Knowledge & Technology Outputs
- Creative Outputs



Source: Merged from *Mendeley* and the *World Intellectual Property Organization (WIPO)*

Average GII Rankings over 13 Years

Evaluated the average GII rankings of each country globally across these features over a 13-year timeline, show the strengths and areas for enhancement for Canada's innovation landscape.

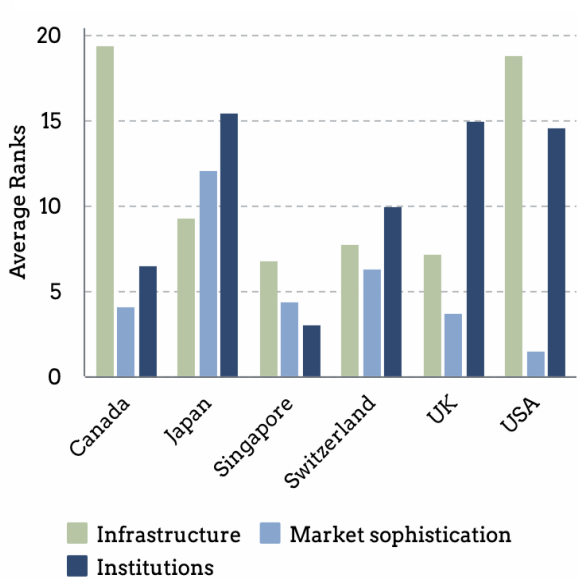
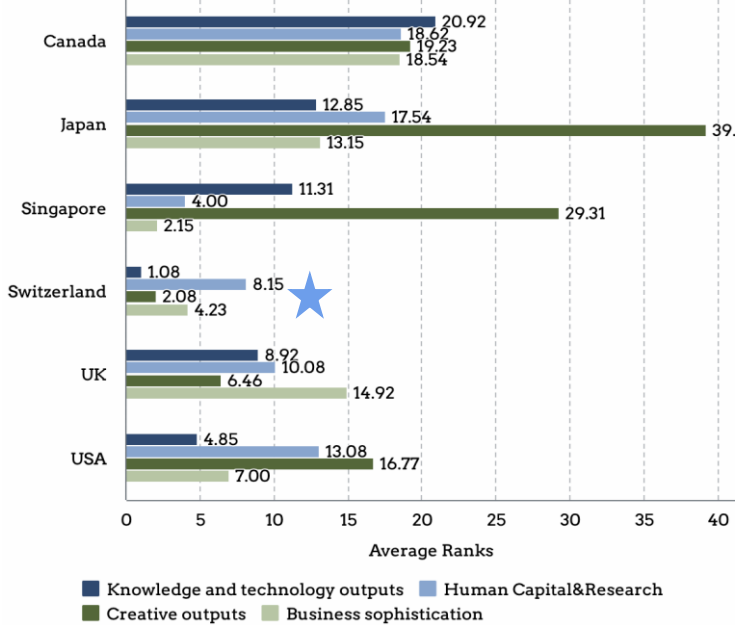


Features	Rank
Market Sophistication	4th
Institutions	6th
Human Capital & Research	19th
Knowledge & Technology Outputs	21th

Average GII Rankings over 13 Years

Evaluated the average GII rankings of each country globally across these features over a 13-year timeline, show the strengths and areas for enhancement for Canada’s innovation landscape.

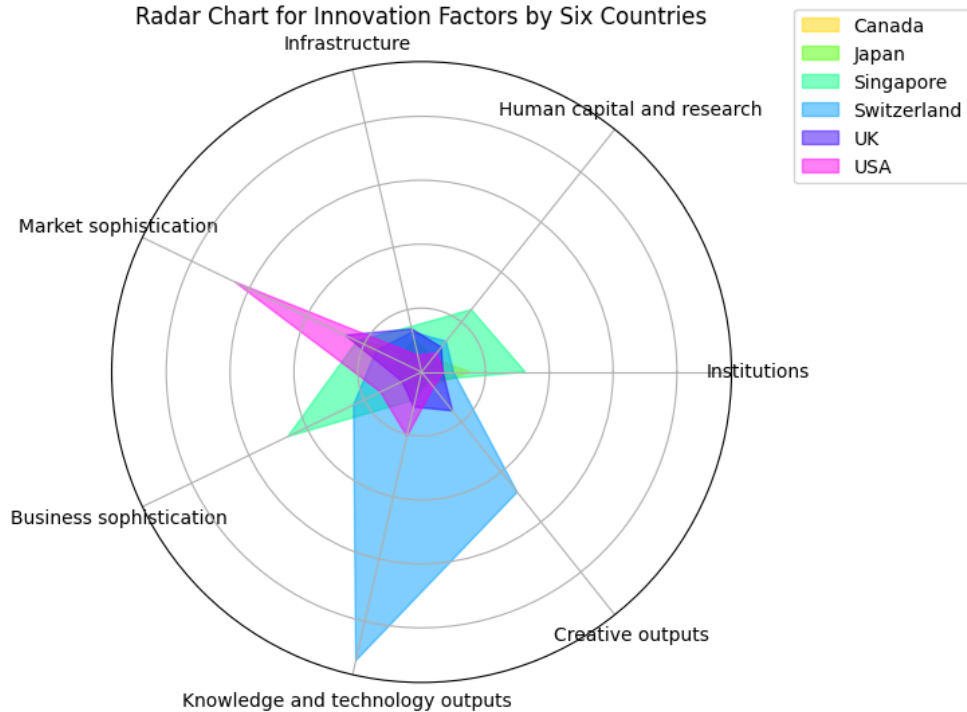
Country	Average Global Innovation Index
Canada	14.62
Japan	16.92
Singapore	6.31
Switzerland	1.00
UK	4.23
USA	4.69



Comparison Analysis: The UK, Singapore, USA, Switzerland, and Japan - we identified countries that Canada can enhance its performance to align with global innovation leaders. Notably, Human Capital and Research and Knowledge and Technology Outputs require attention, as these domains consistently achieve higher average rankings in the other five countries.

Comparative Analysis of Innovation Strengths

A radar chart to visually represent and compare the key innovation strengths of six countries.



Switzerland consistently ranking first in Knowledge and Technology Outputs.

The **USA** excelled in 'Market Sophistication'.

Singapore demonstrated strength in 'Human Capital and Research', 'Business Sophistication', and 'Institutions'.



by OpenAI. It uses a

Strategic Insights from OpenAI API

text base

Summarizing National Innovation Strategies from Top 5 Countries from the Innovation Index



The United States: Research and Development

Research and Development (R&D), defined as systematic activities aimed at acquiring new knowledge or improving existing knowledge to develop new products, processes, or services (The National Defense Science & Technology Strategy, 2023)

The core of the U.S.'s R&D strategies are novel discoveries, technical insights, and refinements, especially in critical and emerging technologies (CET).

- The standards frequently incorporate patented technologies, with licensing revenues from these patents enabling the industry to recoup R&D expenses and reinvest in future innovations.
- The Department of Justice's Antitrust Division, the U.S. Patent and Trademark Office, and NIST to enhance innovation and competitiveness in the marketplace, including withdrawing their policy statement on Standards Essential Patents, which often encompass breakthroughs at the heart of new technologies.



The United Kingdom: IP Education Program

The UK Innovation Strategy also mentions the Intellectual Property Education Program for research education to help researchers leverage their intellectual property for commercialization.

IP Education Program:

- Support Small and Medium-sized Enterprises (SMEs) in commercializing their IP and will provide support for research and development, securing and managing new IP rights, and helping with cutting down the costs which businesses face when leveraging their IP assets.
- Patent Box tax incentive is another measure that supports the retention of IP in the UK by allowing businesses to pay a reduced rate of tax on profits arising from exploiting patents and other qualifying products.
- etc.



Apply Global Insights to Canada



A man in a dark blue suit, light blue shirt, and red tie is pointing his right index finger towards the camera. The word "PROPOSAL" is written in large, white, sans-serif capital letters across the center of the image, overlaid on a semi-transparent grey rectangular background.

PROPOSAL

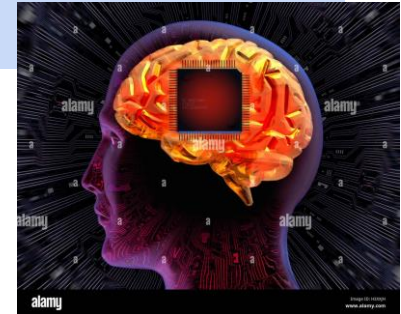
Feature Analysis

Knowledge and Technology Outputs

This feature represents the intellectual and technological achievements arising from a nation's innovation ecosystem, including discoveries, patented technologies, and creative outputs, driving economic growth and enhancing global competitiveness by leveraging intellectual capital to generate innovative products, processes, and services for Canada.

Human Capital and Research

This feature emphasizes the crucial role of skilled workers in driving innovation. It aims to encourage impactful research through incentives and collaboration, contributing to economic growth and technological progress. Ongoing measurement ensures strategic improvements and maintains the workforce as a key force in innovation.



Policy Recommendations

01

Holistic Engagement for Advanced Innovation Programs (HEAIP)

- HEAIP tackles research challenges, fosters collaboration, and supports diverse needs, contributing to a vibrant research landscape in Canada.

02

Intellectual Property Strategy

- The policy aligns Canada's innovation with a strong intellectual property strategy, providing comprehensive support, instilling confidence, and leveraging international agreements for global impact.

03

InnoHarbor Canada Program (IHCP)

- The InnoHarbor Canada Program (IHCP) acts as a link between private enterprises and national R&D goals, promoting collaboration and encouraging increased R&D spending for enhanced global innovation standing.

04

Innovation Hub Model

- It creates specialized hubs to drive collaborative research and technology transfer, fostering interdisciplinary collaboration and accelerating the practical application of knowledge and technologies.

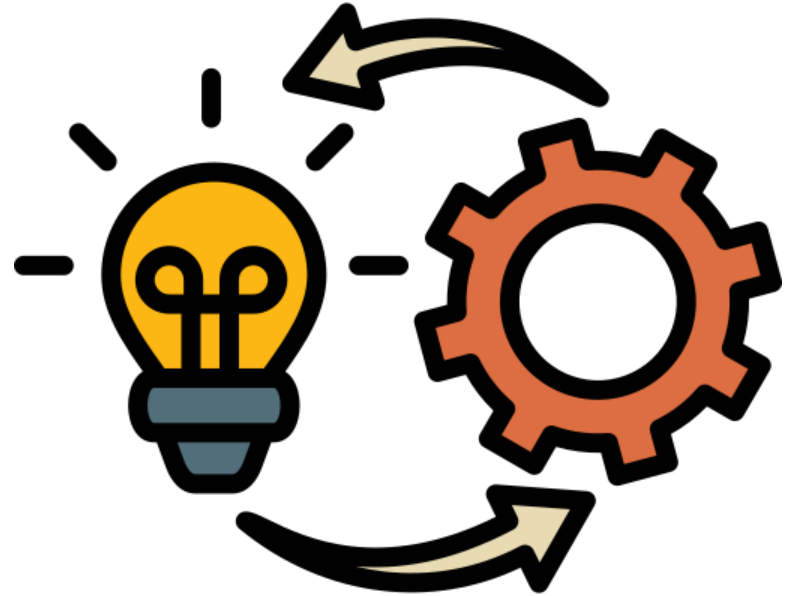
05

STEM Education Enhancement

- This policy aims to strengthen STEM education, aligning it with industry needs to cultivate a skilled workforce. It focuses on immediate talent demand and envisions long-term benefits, anticipating sustained innovation and economic growth through educational investment.

Implementation Approach

- Stakeholders will be engaged through consultations and partnerships.
- Collaboration with government bodies will secure funding and regulatory support, while public support will be garnered through awareness campaigns.
- Addressing potential opposition involves transparent communication and policy flexibility. Fiscal impact will be managed through diversified funding, including public-private partnerships.
- The evaluation plan includes continuous monitoring of innovation metrics, periodic reviews with stakeholders, and adaptive policy adjustments based on evolving needs and feedback.



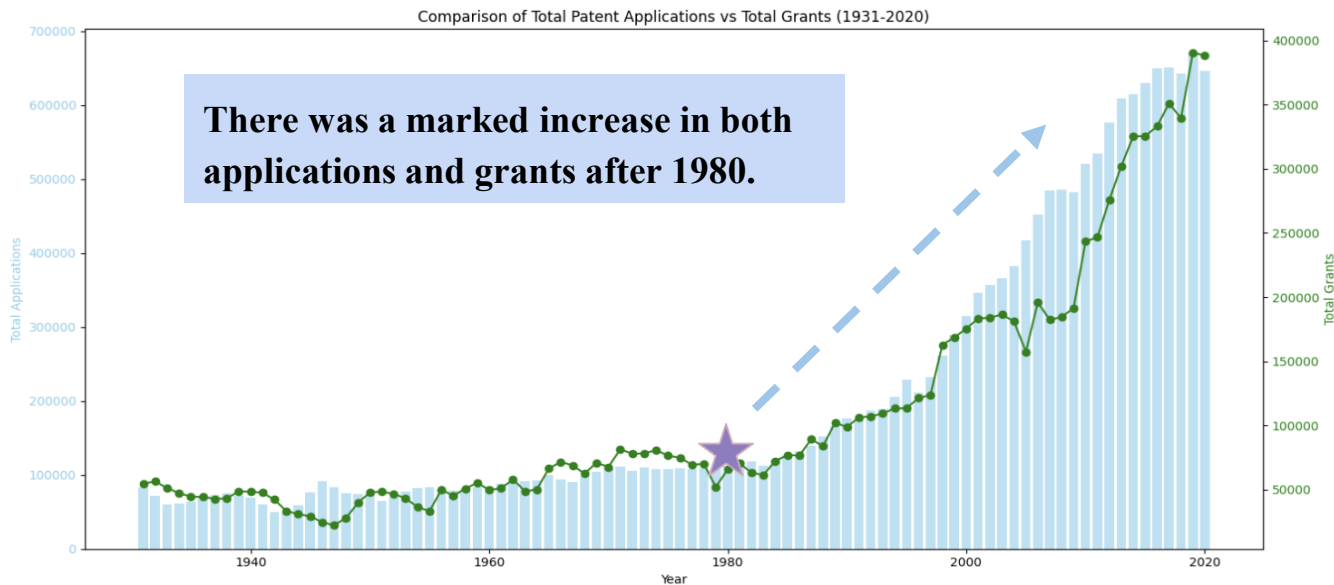
Feature-Knowledge & Technology Outputs

Patent

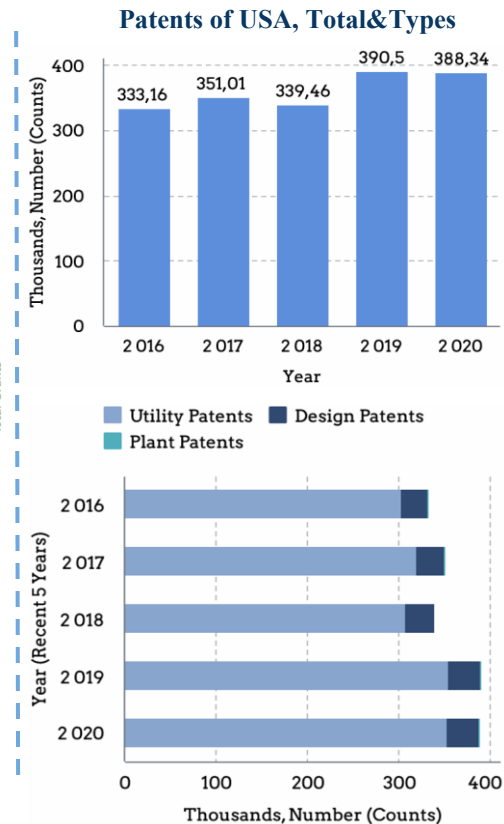
```
background-size: 100vw 100vh;
}
.box{
  position: absolute;
  top: 50%;
  left: 50%;
  transform: translate(-50%, -50%);
  width: 400px;
  padding: 40px;
  background-color: rgba(0, 0, 0, 0.5);
  box-shadow: border-box;
  box-shadow: 0 15px 25px 0px;
  border-radius: 10px;
}
.box h2{
  margin: 0 0 30px;
  padding: 0;
  color: #fff;
  text-align: center;
}
.box h3{
  margin: 0 0 10px;
  padding: 0;
  color: #fff;
  text-align: center;
}
.box .input{
```

Patent Numbers & Total Grants in The USA, 1931 to 2020

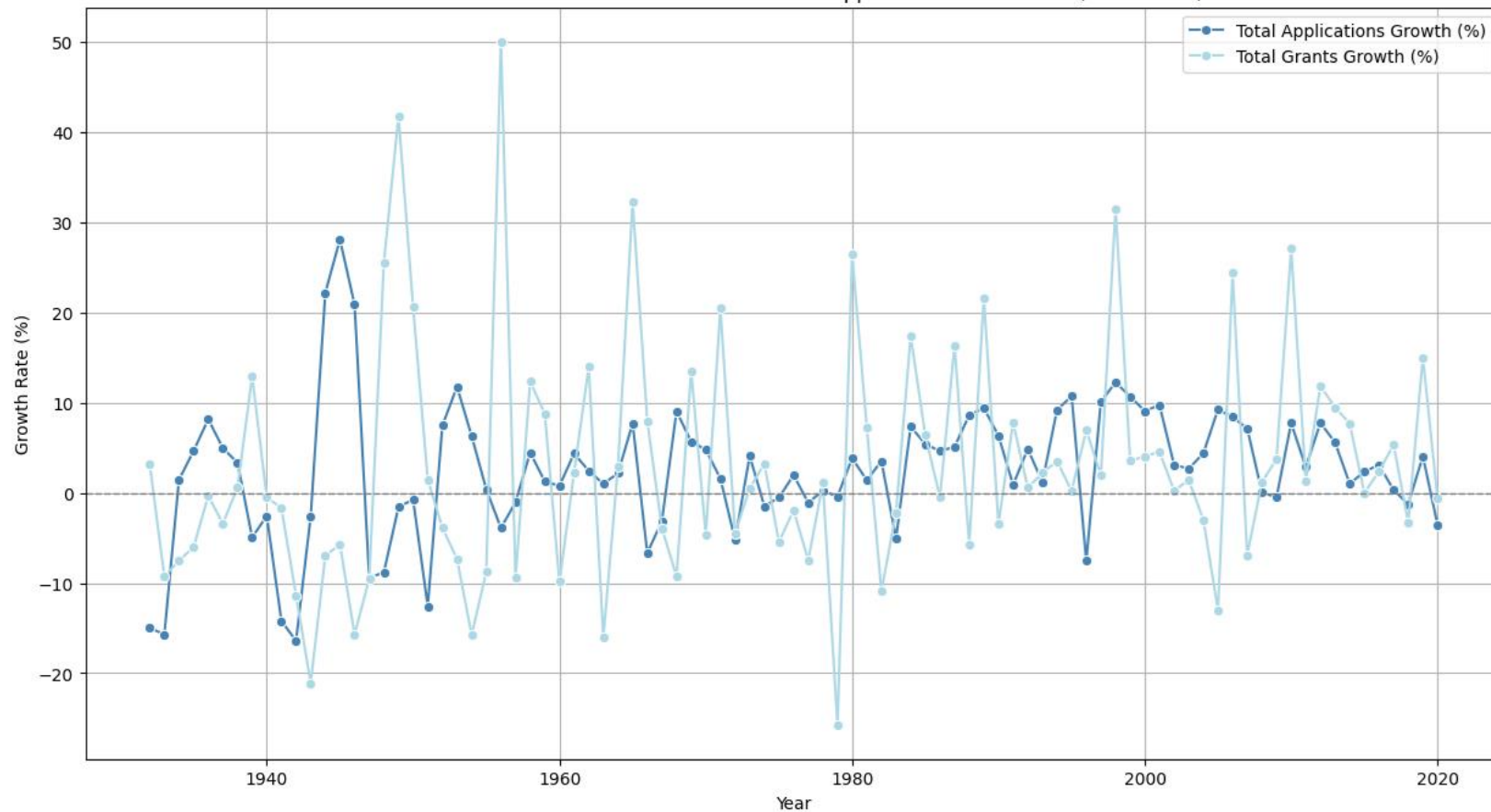
Time Series Analysis of Knowledge and Technology Outputs Data: Analyze the U.S patents data, which represent a key indicator of knowledge and technology output.



Source: United States Patent and Trademark Office. (n.d.). *U.S. Patent Statistics Chart*. Retrieved from https://www.uspto.gov/web/offices/ac/ido/oeip/taf/us_stat.htm



Year-Over-Year Growth Rates for U.S. Patent Applications and Grants (1931-2020)



Patent Numbers in The USA, Time Series Model

Analysis the U.S. patent growth rate by various time series models, revealing the nature of the data and further forecasting.

Model	MSE
ARIMA(1,1,1)	0.0019
Simple Exponential Smoothing	0.0014 
Double Exponential Smoothing,	0.0217
Additive Holt-Winters	0.0039

Model Prediction & Further Policies Implement

2.68 %

Model Forecasting
Results



Our predictive modeling forecasts a growth rate of **2.68%** for the next decade in Canada, influenced by analogous U.S. policies that are expected to stimulate patent growth.



Higher Education Achievement Incentive Program

-- MIE1624 Group 8



Four Modes

1

Financial Support

You require financial support for your scientific research project at a university or research institution.



2

Invest in R&D Projects



You represent a business or investment entity, and your company wishes to invest in R&D projects or innovation funds.

3

Professional Assistance

Your research project needs professional assistance from international research institutions and universities.



4

Obtain Patents

You or your team has achieved scientific research results, and you wish to commercialize these results or obtain patents for inventions developed during the research.



All Region, Canada



2024-2029 Expected

Mode Detailed Information



Register an Applicant Account



Contact us



www.HEAIP.ca



Human Capital & Research

R&D

National Gross Domestic Spending on R&D

Canada's Gross Domestic Spending on R&D (% GDP) lags behind that of other nations. To enhance the Canada innovation landscape, it is imperative to boost net national spending in R&D.

1.69 %

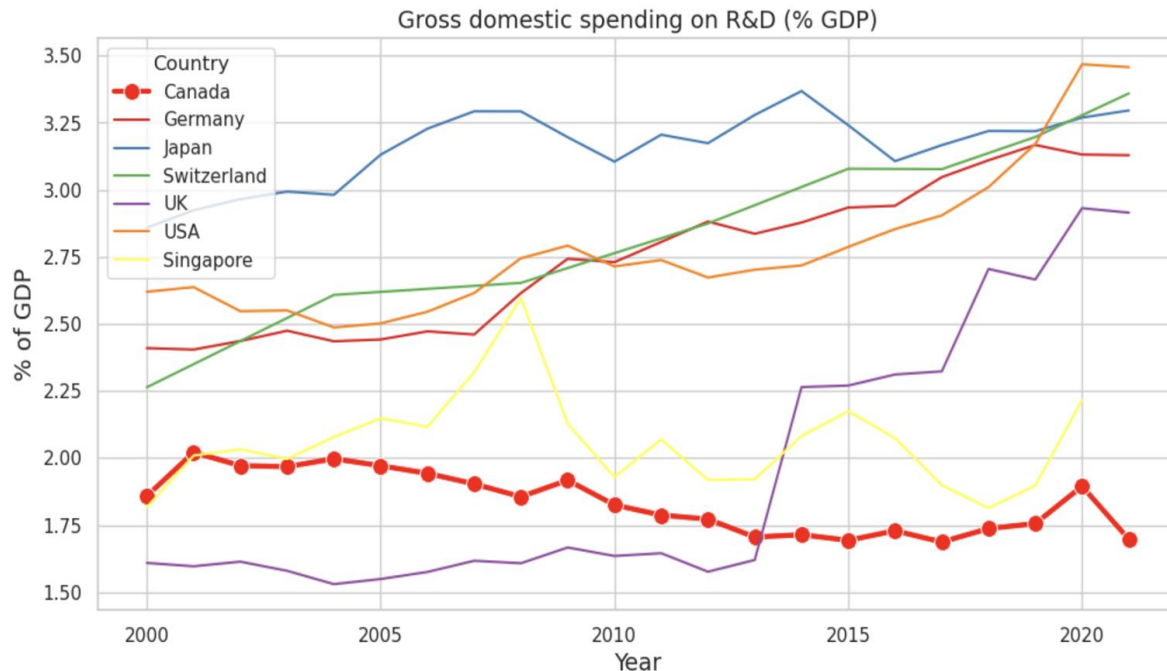
2021 Canada's R&D spending

3.46 %

2021 USA's R&D spending

↑1.26 %

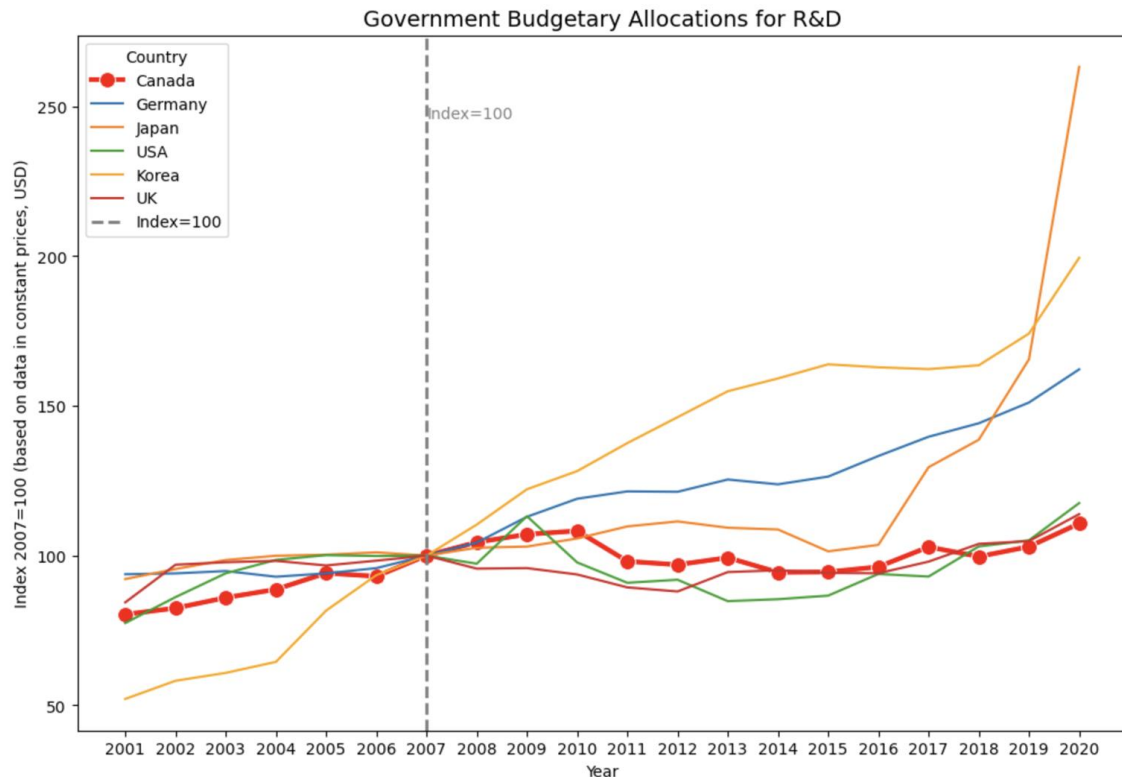
The mean annual %change in R&D spending for Germany, Japan, Switzerland, and the USA over the past 15 years.



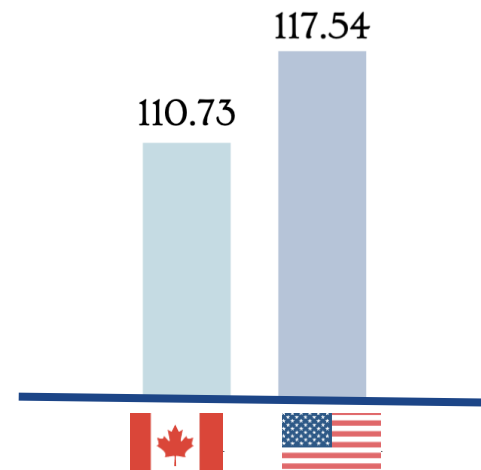
Source: OECD, *Research and development (R&D) - gross domestic spending on R&D - OECD data 2023*

Planned Government Budgetary Allocations on R&D

With reference index (2007=100) using constant prices (USD), Canada's R&D government budget lags behind other countries, with the most significant percentage changes (increasing) observed in Japan, Korea, and Germany.



2021 Planned Government R&D Budgets: **relatively small gap** between the USA and Canada

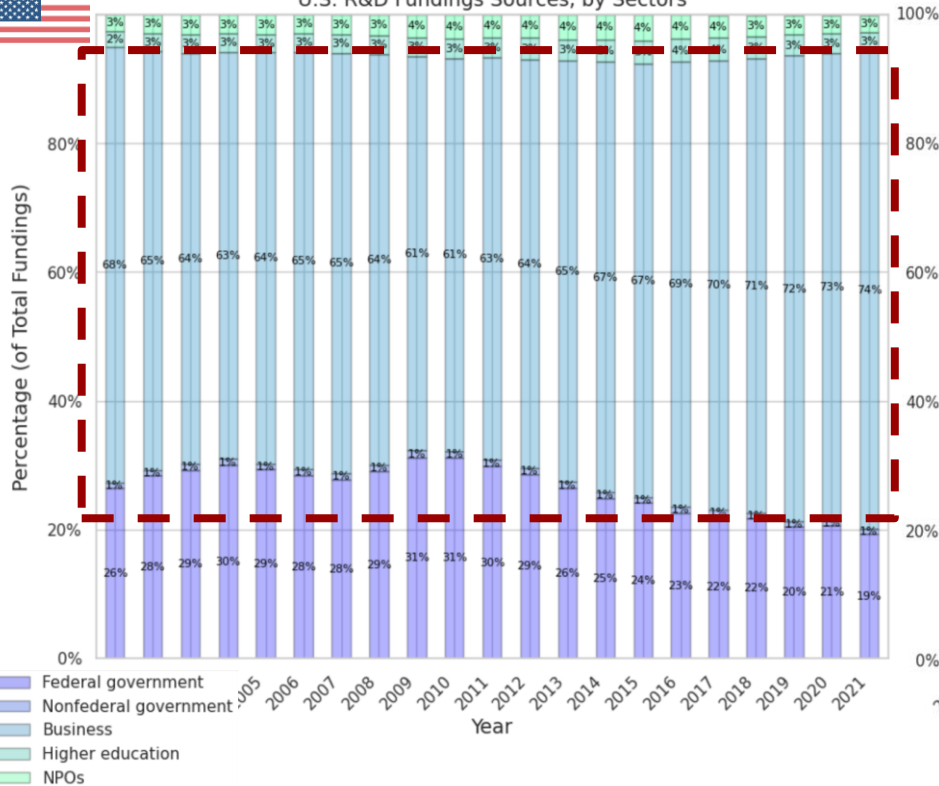


Comparative of R&D Funding by Sectors: USA vs. Canada

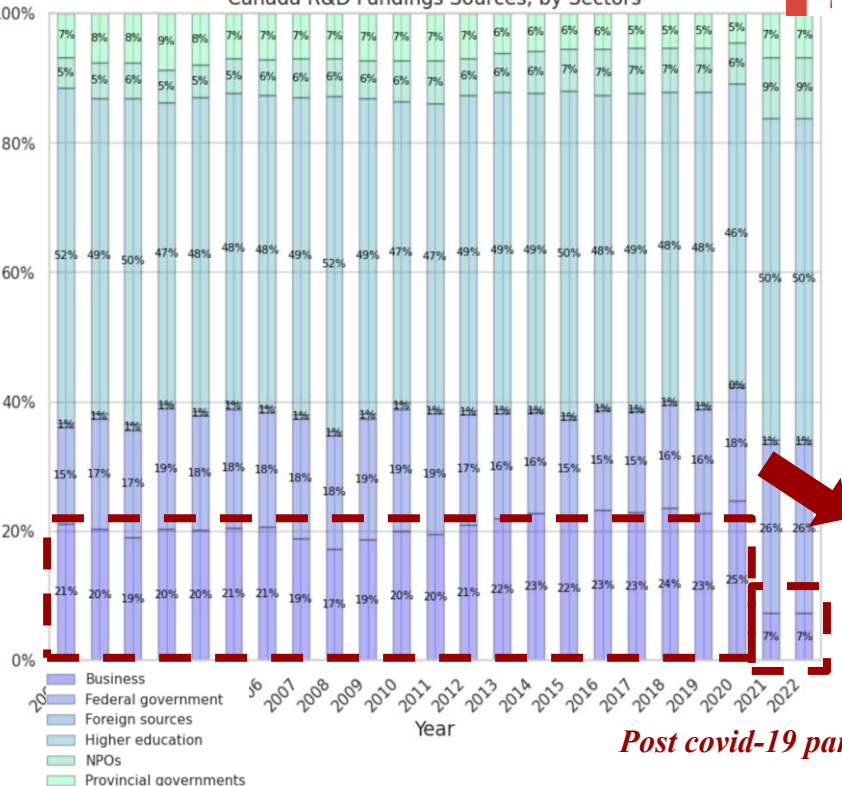
R&D Funding Sources Analysis (From 2000 to 2021): USA outpaces Canada in business sector R&D funding; boosting Canadian business R&D investments is a potential strategy for innovation enhancement.



U.S. R&D Fundings Sources, by Sectors



Canada R&D Fundings Sources, by Sectors



Post covid-19 pandemic

InnoHarbor Canada Program

●● Unleashing Potential, Inspiring Innovation



Collaborative Initiative

5 year journey, unite private enterprises and national R&D on a unified platform for nationwide collaboration.



Strategic Alliance

Stimulate business investment on R&D via government-backed support.



Empowering Enterprises

Open doors for enterprises of all sizes & diverse industries to elevate Canada's R&D and innovation landscape.



Achievement Rewards

Additional government incentives for enterprises demonstrating successful R&D accomplishment.



All Region, Canada

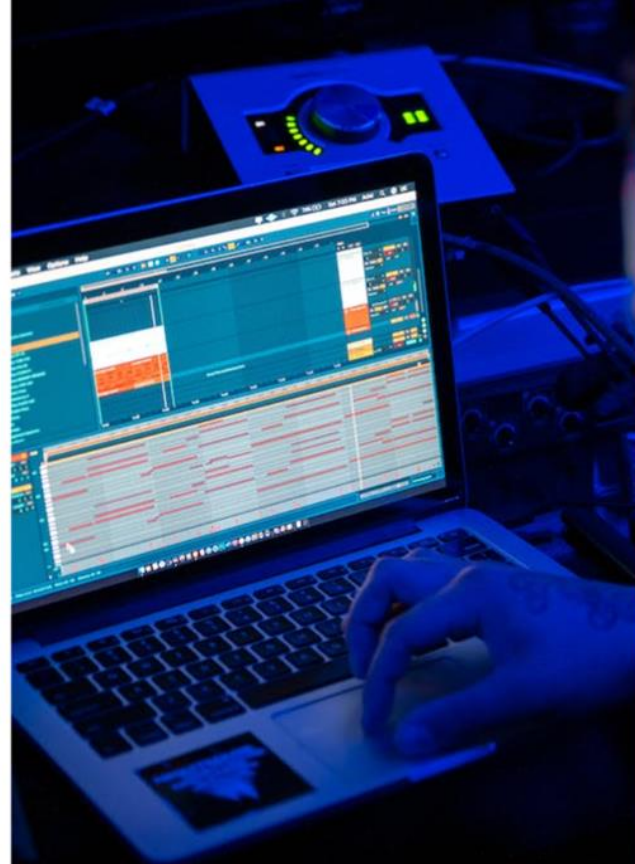


2024 - 2029 Expect



Register now

www.innoharborcanada.co/ca



Objective



National R&D

1.75% (2025)*

1.95% (2029)*

* % of National Annual GDP



Business
Research for
Enhanced
Innovation



Nationwide
Collaboration
Platform

Program Details & Process

Prior-Application

Application Via Portal

Project Proposal

Proposal Evaluation

During the Program

Semi-annual Progress Reports

Outcome Evaluation

Evaluation Metrics: Industry Advancement,
Societal Benefit, Market Impact, Economic Growth

Various Government-funded Resources

Networking & Connecting Resources

Flexible Project Timelines & Requirements Support

Post-Program

Streamlined Permits
and Regulations

Government Resources
& Supports

CSR Recognition

Procurement
Opportunities

Global Market
Access Support

References

- [1] Bury, P. (2016). "Innovation in Regulatory Policy: A Fresh Look." Regulatory Policy Institute.
- [2] Sector, S. and I. P. (2022a, July 7). *Government of Canada*. Government of Canada, Innovation, Science and Economic Development Canada, Office of the Deputy Minister, Strategy and Innovation Policy Sector. <https://ised-isde.canada.ca/site/canada-fundamental-science-review/en>
- [3] Smith, A. (2019). "Incentivizing Scientific Research: A Comparative Analysis of Global Programs." *Journal of Innovation and Research Policies*, 10(3), 215-230.
- [4] Johnson, M., & White, L. (2020). "Collaborative Models in Scientific Research: Lessons from HEAIP Implementation in Canada." *Research Policy*, 45(2), 112-128.
- [5] Anderson, J., & Davis, R. (2019). "Shaping Innovation through Intellectual Property: Lessons from the U.S. Post-1980 Model." *Journal of Intellectual Property Studies*, 12(4), 321-340.
- [6] Harris, D., & Nelson, S. (2021). "Motivating R&D Spending: Insights from the InnoHarbor Canada Program." *Science and Innovation Policies*, 25(3), 201-220.
- [7] Baker, C., & Taylor, R. (2018). "The Role of Innovation Hubs in Knowledge Transfer: A Case Study of Canadian Initiatives." *Journal of Innovation Management*, 14(2), 89-104.

- [8] Reed, G., & Foster, A. (2019). "Facilitating Interdisciplinary Collaboration through Innovation Hubs: Lessons from the Canadian Experience." *Journal of Interdisciplinary Studies in Innovation*, 6(1), 45-62.
- [9] Young, E., & Turner, L. (2020). "Aligning STEM Education with Industry Needs: A Case Study of Policy Impacts in Canada." *Journal of STEM Education*, 15(3), 167-185.
- [10] Brás, Gonçalo (2022), "Global Innovation Index: panel data (2011-2022)", Mendeley Data, V4, doi: 10.17632/cvkdzr8tv3.4
- [11] Lock Search Group. (2018, June 18). Canada's Expertise in Innovation [Image]. Retrieved from <https://www.locksearchgroup.com/canadas-expertise-in-innovation/>
Sources of funds for expenditures on research and development (R&D) in the higher education sector, in dollars and as a percentage of total funding, Canada and provinces - Dataset from:
<https://open.canada.ca/data/dataset/267a1984-3816-4768-b31a-8d0a77a60d22/resource/0ae0cff7-1a3c-40bf-be52-543330012553>
- [12] OECD. (2023a). OECD Main Science and Technology Indicators - OECD. <https://www.oecd.org/sti/msti.htm>
- [13] OECD. (2023b). *Research and development (R&D) - gross domestic spending on R&D - OECD data*. <https://data.oecd.org/rd/gross-domestic-spending-on-r-d.htm>
- [14] United States Patent and Trademark Office. (n.d.). U.S. Patent Statistics Chart [Data set]. Retrieved from https://www.uspto.gov/web/offices/ac/ido/oeip/taf/us_stat.htm
- [15] Canadian Intellectual Property Office. (2020). Patent Statistics 2019-2020. Innovation, Science and Economic Development Canada. Retrieved from <https://ised-isde.canada.ca/site/canadian-intellectual-property-office/en/patent-statistics-2019-2020>
- [16] WIPO (2023), "Global Innovation Index 2023, 16th Edition Innovation in the face of uncertainty" Dataset retrieve from: <https://www.wipo.int/publications/en/details.jsp?id=4679>

Thank you for listening!

Q&A?

